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PROFESSIONAL SUMMARY

Senior Hardware Engineer specializing in power electronics (SMPS, LLC, Flyback) with experience leading designs from concept to mass production in consumer and industrial products. Proven track record in cost optimization, EMI compliance, and international vendor validation. Master's researcher focusing on AI-based battery SOH estimation for EV applications.

Technical SKILLS

- Power Electronics: Flyback, LLC, PFC, Three-Phase SMPS
- Magnetics Design: Core selection, gapping, winding optimization
- PCB Design: High-speed, impedance control, EMI/EMC, thermal management.
- Simulation: LTspice, SIMPLIS, MATLAB/Simulink
- Tools: Altium Designer, AutoCAD
- Programming: Python (automation, data analysis, test scripting), C (embedded systems fundamentals).
- Automated Instrument Control using PyVISA (SCPI-based communication)
- Test & Measurement / Instrumentation: Oscilloscope, Function Generator, Electronic Load, Spectrum Analyzer, Power Analyzer
- Embedded Systems: Zynq SoC, DDR3, Ethernet, USB
- Wire Harness for TV cables using AutoCad

EXPERIENCE

Sr. R&D HW Electronics Engineer

Pylon

Oct 2025 – Present

- Developing a Battery Management System (BMS) for LFP battery packs using Texas Instruments AFE and fuel gauge ICs.
- Implementing cell monitoring, protection mechanisms (OV/UV/OC), and battery state estimation (SOC/SOH).
- Leading the design of SMPS for three-phase Smart Electric Meters.
- Reduced BOM cost by approximately 40% compared to the previous generation design.
- Designed EMI filter architectures achieving compliance with CISPR/IEC standards for three-phase smart meter SMPS.
- Designed automated production test jigs reducing validation time and improving factory throughput.

Sr. R&D HW Electronics Engineer

El-Araby Group TV

Jan 2024 – Oct 2025

- Global Supply Chain & Validation: Executed on-site validation testing and vendor technical audits during two business residencies in China, ensuring component quality and boards validation tests for mass production.
- Full-Lifecycle Ownership: Directed the hardware development process from initial schematic capture and component selection to final assembly
- PCB validation and qualification for mass production.

R&D HW Electronics Engineer

El-Araby Group TV

June 2022 – Jan 2024

- Designed and engineered a 75W quasi-resonant flyback converter.
- Co-designed a 150W dual-flyback power supply with parallel converters.
- Performed analytical design calculations for LLC resonant converters.
- Ensured compliance with IPC PCB standards and IEC regulatory requirements.

Early Training Experience

Alexandria Petroleum Maintenance CO, Abu Qir Fertilizers CO, Alexandria Electricity Distribution CO

Aug 2018 – Aug 2019

- Shadowed field technicians on preventive and corrective maintenance, gaining a practical view of industrial operations, safety protocols, and workflow management in a high-stakes petroleum environment.
- Exposure to industrial operations in petroleum, chemical, and electrical distribution sectors

PROJECT

Battery Management System (BMS) – In Progress

Pylon

- Architecting a full BMS solution for LFP battery packs including sensing, protection, and balancing strategy.
- Integrating TI monitoring and gauging ICs with system-level control and validation workflows.

Three-Phase SMPS

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- Designed cost-optimized three-phase SMPS with robust protection against grid transients.
- Collaborated with IC vendors for topology optimization and component selection.

150W Dual Flyback

Elaraby Group

- Part of a team responsible to design a 150W power supply for a television,
- Implemented two parallel flyback converters to power the LED backlight and main board independently.
- Integrated an input PFC boost stage to meet efficiency standards and managed the full development cycle from concept to validation.

75W Flyback

Elaraby Group

- Spearheaded the development of the company's first in-house SMPS, a 75W flyback converter.
- Responsible for the full magnetic design, including core selection, gapping, and winding structure for the custom transformer, establishing a new internal capability.

ZettBrett EVB - InProgress

Phill's Lab Course Graduation Project

- This is a 10-Layer impedance-controlled, double-sided assembly, 400+ parts (down to 0201 passives)
- Features:
 - 1) AMD/Xilinx Zynq System-on-Chip (SoC) ('half APU, half FPGA')
 - 2) 1GB DDR3 Memory
 - 3) 4GB eMMC Memory
 - 4) Gigabit Ethernet
 - 5) USB HS OTG
 - 6) On-board USB-to-JTAG
 - 7) Embedded Linux-capable

EDUCATION

M.Sc. in Power Electronics (In Progress)

University of Cairo – Egypt

Research Point: SOH Estimation in Electric Vehicles using AI

Relevant Coursework: Advanced Power Electronics, Magnetics Design, Adaptive Control, Intelligent Control

Bachelor of Science in Electrical Engineering

University of Alexandria – Egypt • Power Department • 2020 – With Grade Distinct with honor (fifth)

Extracurricular Activities

- Participated in multiple robotics competitions including Line Maze Robot (LMR), Minesweeper Robot, and Robocon.
- Designed and tested embedded control systems, sensors integration, and real-time decision algorithms.
- Gained hands-on experience in system debugging, teamwork under competition constraints, and rapid prototyping.

Professional Development

- Advanced Hardware Design – Phill's Lab
- PCB Materials and Stack-Up – FEDEVEL
- PCB Thermal Management – Udemy
- PCB Design & Manufacturing – Udemy
- PLC Level 1 – EGYCET

Soft SKILLS & Additional Information

Soft Skills: Leadership, Technical Communication, Cross-Functional Collaboration, Value Engineering Mindset, Product Development Support, Teamwork.

Professional Development: Management & Project Management courses by McKinsey & Company, Pluralsight, and Coursera (Google Project Management – In Progress); Entrepreneurship Program – GYP.

Languages: Arabic (Native), English (Fluent), Chinese (HSK1 – Basic), Japanese (JLPT N5 – Basic).

Interests: Language learning, Entrepreneurship, Reading.